$\boxed{L6}$ Evaluating the OLS fit

Today's agenda:

- R - squared
- Estimating standard errors
- Hypothesis testing

R-squared / Coefficient of Determination

Recall:

- $Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$ ~ vector of training outputs

- $\hat{Y} = \begin{pmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{pmatrix}$ ~ vector of predicted outputs under OLS

  $= HY$ (assuming X has full col. rank)

- $RSS = \|Y - \hat{Y}\|^2$

  $= \sum_1^n (y_i - \hat{y}_i)^2$

Define:

a) $\underbrace{TSS}_{\text{total sum of squares}} = \sum_1^n (y_i - \bar{y})^2$, where $\bar{y} = \frac{1}{n}\sum_1^n y_i$

$= \|Y - \bar{Y}\|^2$, where $\bar{Y} = \begin{pmatrix} \bar{y} \\ \bar{y} \\ \vdots \\ \bar{y} \end{pmatrix}$

Note: TSS captures the <u>inherent</u> variability of training outputs

b) $\underbrace{MSS}_{\text{Model sum of squares}} = \sum_{1}^{n} (\hat{y}_i - \bar{y})^2$

$$= \| \hat{Y} - \bar{Y} \|^2$$

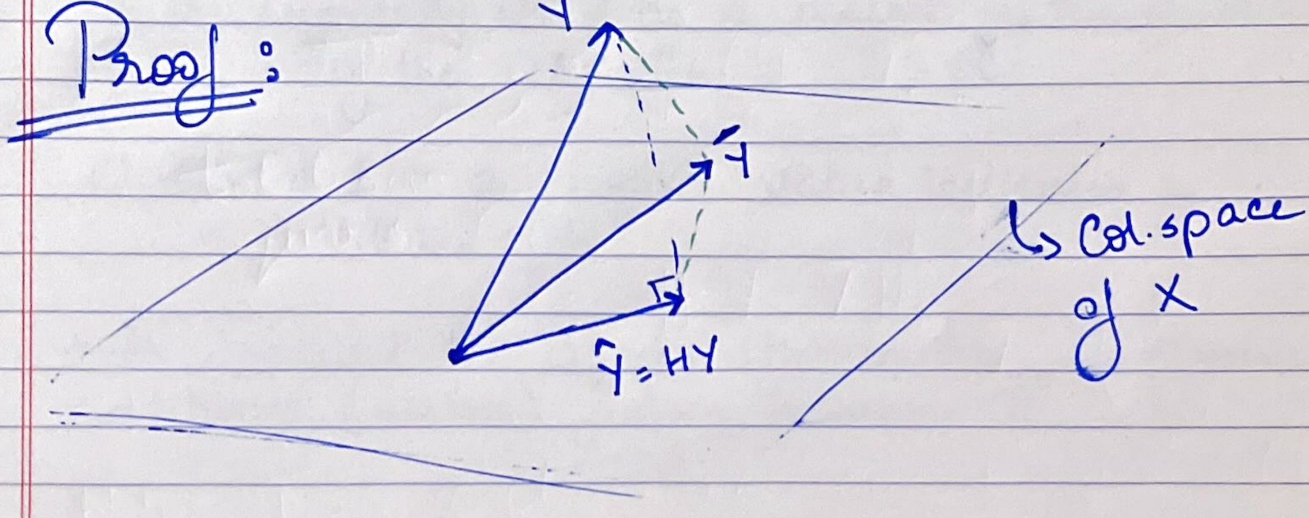
Note: MSS captures the o/p variability explained by the model

Note: In some references, MSS is referred to as ESS (explained sum of squares)

Turns out:

$$\underbrace{TSS}_{\text{o/p variability}} = \underbrace{MSS}_{\text{o/p variability explained by (OLS) model}} + \underbrace{RSS}_{\text{mismatch between o/p \& model}}$$

Proof:



- $Y-\hat{Y}$ is $\perp$ to col. space of X
- $\bar{Y}$ lies in col. space of X (Why?)
- By Pythagoras theorem,

$$\underbrace{\|Y-\bar{Y}\|^2}_{TSS} = \underbrace{\|Y-\hat{Y}\|^2}_{RSS} + \underbrace{\|\hat{Y}-\bar{Y}\|^2}_{MSS}$$

$\square$

We define

$$R^2 = \frac{MSS}{TSS} = 1 - \frac{RSS}{TSS}$$

R-squared
coefficient of determination

R^2 is typically used as a metric of the quality of the OLS fit.

$0 \le R^2 \le 1$; a higher value indicates a better fit.

★ Should be used with caution. Introducing additional (useless) features increases R^2.

Estimating standard errors

Under our statistical model, we assumed

$$y_i = \beta_0^* + \sum_{1}^{p} \beta_j^* x_i(j) + \underbrace{\varepsilon_i}_{\mathcal{N}(0,\sigma^2)}$$

$$= \cancel{\beta^*}$$

$$= \left\langle \beta^*, \begin{pmatrix} 1 \\ x_i \end{pmatrix} \right\rangle + \varepsilon_i$$

Here β^* & σ^2 are unknowns.

Turns out:
- $\hat{\beta}$ is MLE of β^*
- $E[\hat{\beta}] = \beta^*$

  $\Rightarrow$ MLE is <u>unbiased</u>
- $Cov(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$

Q: Can MLE also estimate σ^2?

A: Yes!

$$\hat{\sigma}^2 = \frac{RSS}{n-p-1}$$

↳ unbiased estimate of σ^2

We can use these facts to develop confidence intervals on the regression co-efficients.

Note:

$$\underbrace{Var(\hat{\beta}_i)}_{\text{coefficient of } i^{th} \text{ feature}} = \sigma^2 \underbrace{C_i}_{i^{th} \text{ diagonal entry of } (X^T X)^{-1}}$$

$$\Rightarrow \quad Var(\hat{\beta}_i) \approx \hat{\sigma}^2 C_i$$

Define $\underbrace{\underset{\text{standard error}}{SE}(\hat{\beta}_i) = \sqrt{\hat{\sigma}^2 C_i}}_{\text{estimate of std. deviation of } \hat{\beta}_i}$

estimate of ~~variance~~ std. deviation of $\hat{\beta}_i$

$$\Rightarrow \boxed{\text{Can assume } \beta_i^* \text{ lies within } \hat{\beta}_i \pm SE(\hat{\beta}_i)}$$

Captures our uncertainty in the estimation of β_i^*

Hypothesis testing

- A rigourous statistical framework to answer yes-no questions.

  Eg Does smoking increase likelihood of lung ~~cancer~~ cancer?

  Eg Is a certain regression coefficient zero?

  Eg Is the avg. blood pressure among mice in the treatment group the same as that in the control group?

4 steps

Step 1. Define null & alternative hypotheses

- Null hypothesis H_0 represents our default state of ~~bel~~ belief.

- Alternative hypothesis : something different/unexpected surprising; Complement of H_0.

Goal : See if data allows us to confidently reject the null hypothesis.

Note: This setup is assymetric between H_0 & H_a.

- Rejecting H_0 means we are confident H_0 does not hold.

- Failure to reject H_0 $\Rightarrow$ we are unable to reject H_0, based on available data. Either H_0 does hold, or we do not have enough data to reject H_0.

Step 2. Design the test statistic

The test statistic summarizes the extent to which the data is consistent with H_0

Step 3 Compute p-value

The p-value is the probability of ~~that the~~ observing a test statistic at least as extreme as the observed value.

Small p-value $\Rightarrow$ evidence against H_0

Step 4 Decide whether or not to reject null hypothesis.

Typically, we reject H_0 if p-value $< \underline{0.05}$.

threshold is a hyperparameter
could be even smaller for mission critical applications

Application to linear regression

Q: Is the i^{th} feature useful in predicting the o/p?

Eg: Does gender play a role in predicting loan default probability?

This is posed as a hypothesis testing problem as follows:

$$\left.\begin{aligned} H_0 &: \beta_i^* = 0 \\ H_a &: \beta_i^* \neq 0 \end{aligned}\right\} \text{Step 1}$$

Turns out:

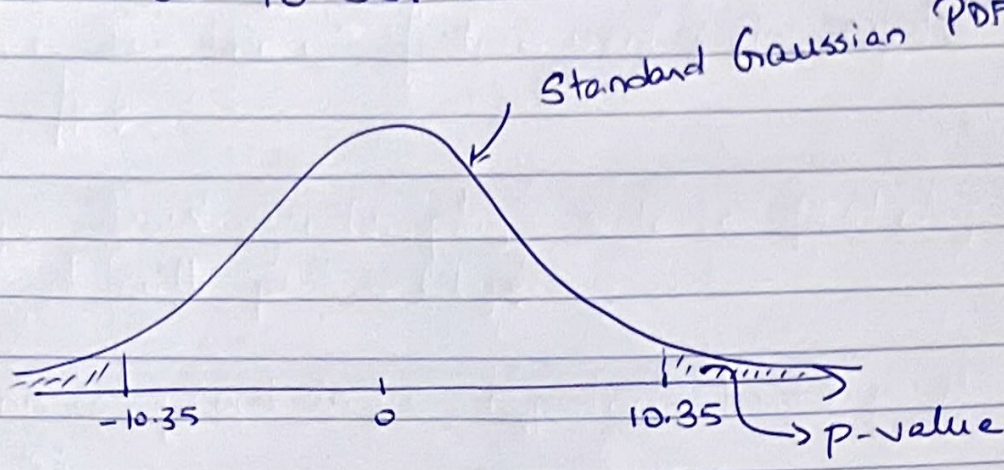
$$\frac{\hat{\beta}_i}{\sqrt{C_i \sigma^2}} \sim \mathcal{N}(0,1)$$

C_i: i^{th} diagonal entry of $(X^T X)^{-1}$

But we don't know σ^2!

Turns out: $$\left.\frac{\hat{\beta}_i}{\sqrt{C_i \hat{\sigma}^2}} \approx \mathcal{N}(0,1)\right\} \text{Step 2}$$ test statistic

Eg Say that the value of the test statistic evaluates to 10.35.



$$\text{p-value} = 4 \times 10^{-25}$$

Moral: If H_0 holds, the probability of this extreme a test statistic is very unlikely.

$\Rightarrow$ We may reject the null hypothesis.

☆ Give example via advertising dataset.